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Exam Code	CS305E
Exam Name	AI Powered CI/CD

GitHub URL : <https://github.com/Subrat-Dwivedi-22/github-skills-challenge>

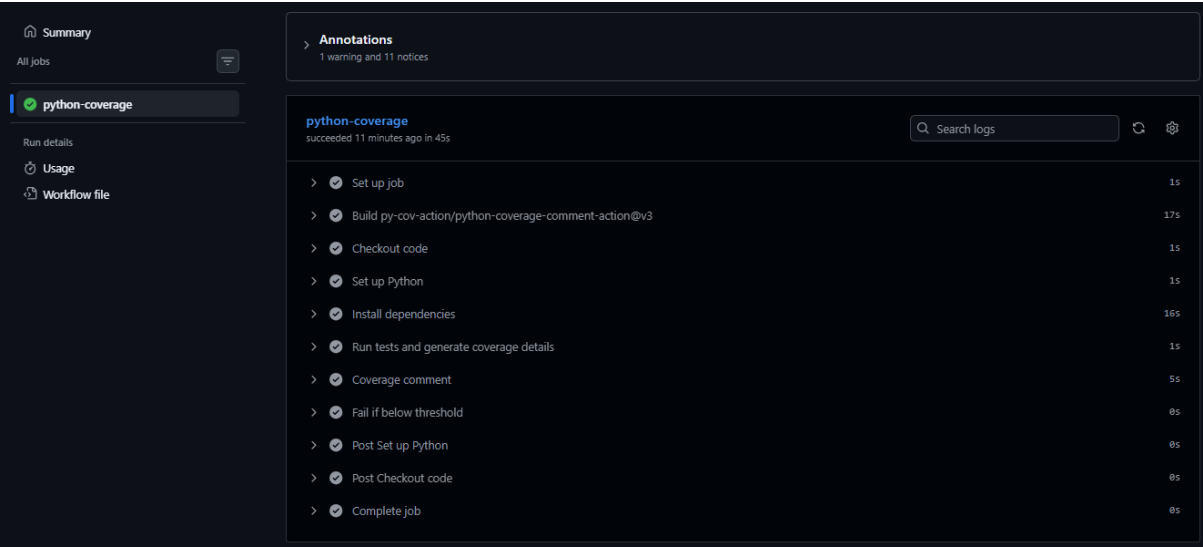
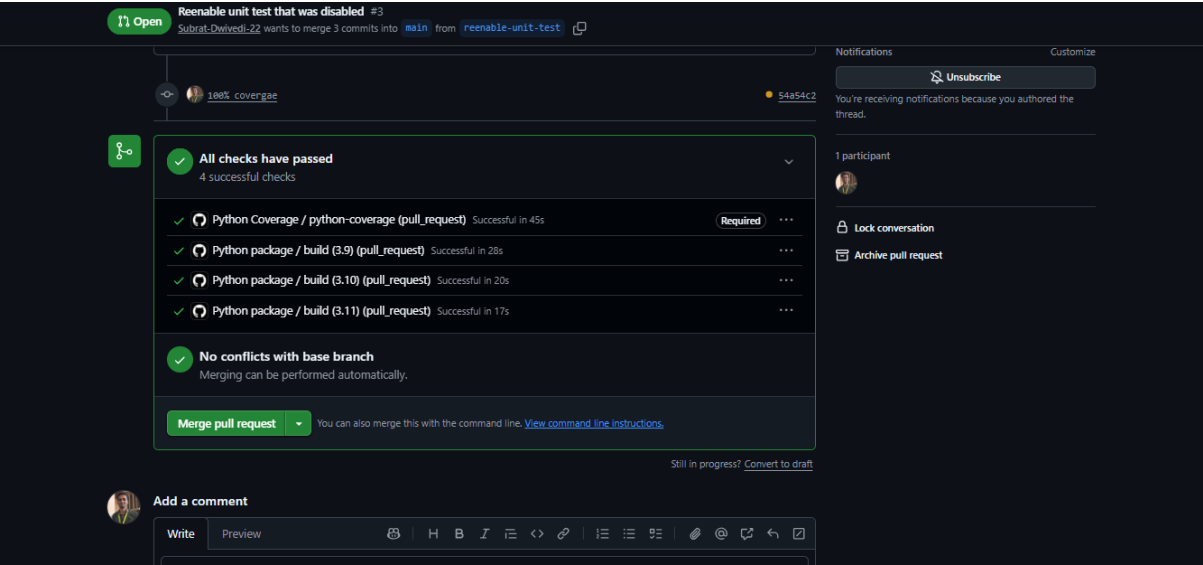
Pull Request URL : <https://github.com/Subrat-Dwivedi-22/github-skills-challenge/pull/3>

Screenshot showing the GitHub Actions workflows

The screenshot displays the GitHub Actions interface. On the left, the 'Actions' sidebar is visible with a 'New workflow' button and a list of workflow categories: Python Coverage, Python package, Step 1, Step 3, Step 4, Management, Caches, Attestations, Runners, Usage metrics, and Performance metrics. The main area shows '7 workflow runs' for the 'reenable-unit-test' workflow. The runs are listed in a table with columns for Workflow, Event, Status, Branch, and Actor. The first three runs are successful (green checkmark) and the last four are failed (red X). The failed runs are labeled 'Reenable unit test that was disabled' and 'Exercise: Test with Actions'.

Workflow	Event	Status	Branch	Actor
reenable-unit-test	Python Coverage #3: Pull request #3 synchronize by Subrat-Dwivedi-22	Success	main	Subrat-Dwivedi-22
reenable-unit-test	Python package #3: Pull request #3 synchronize by Subrat-Dwivedi-22	Success	main	Subrat-Dwivedi-22
reenable-unit-test	Python package #2: Pull request #3 synchronize by Subrat-Dwivedi-22	Success	main	Subrat-Dwivedi-22
reenable-unit-test	Python Coverage #2: Pull request #3 synchronize by Subrat-Dwivedi-22	Failed	main	Subrat-Dwivedi-22
reenable-unit-test	Python package #1: Pull request #3 opened by Subrat-Dwivedi-22	Failed	main	Subrat-Dwivedi-22
reenable-unit-test	Python Coverage #1: Pull request #3 opened by Subrat-Dwivedi-22	Failed	main	Subrat-Dwivedi-22
reenable-unit-test	Exercise: Test with Actions Step 1 #1: Issue comment #2 (comment) created by Subrat-Dwivedi-22	Failed	main	Subrat-Dwivedi-22

Screenshot showing a successful test workflow execution



Screenshot showing the coverage result/comment on the pull request

This screenshot shows a GitHub pull request titled "Reenable unit test that was disabled" (#3) from the "reenable-unit-test" branch. The pull request is merged. A comment from the "github-actions" bot, posted 23 minutes ago, displays a "Coverage report". The report shows "Coverage 100%" and "PR Coverage 100%". Below the report, it states: "This PR does not seem to contain any modification to coverable code." The right sidebar shows the pull request's metadata, including "No reviews", "No one assigned", and "No labels".

This screenshot shows the same GitHub pull request, but now it is open. The status bar at the top indicates "All checks have passed" with 4 successful checks. The checks listed are: "Python Coverage / python-coverage (pull_request)" (Successful in 45s, Required), "Python package / build (3.9) (pull_request)" (Successful in 28s), "Python package / build (3.10) (pull_request)" (Successful in 20s), and "Python package / build (3.11) (pull_request)" (Successful in 17s). Below the checks, it states "No conflicts with base branch" and "Merging can be performed automatically." A "Merge pull request" button is visible. The right sidebar shows the pull request's metadata, including "No reviews", "No one assigned", and "No labels".

Screenshot showing the final repository/workflow files

This screenshot shows the GitHub repository for 'github-skills-challenge' at the path '/github/workflows/'. The left sidebar displays the file tree with the 'workflows' directory expanded, showing files from '0-start-exercise.yml' to 'python-package.yml.example'. The main content area shows a commit by 'Subrat Dwivedi-22' titled 'Added Coverage Workflow(Task 4)' from 44 minutes ago. Below the commit information is a table listing the workflow files and their commit history.

Name	Last commit message	Last commit date
..		
0-start-exercise.yml	Initial commit	10 hours ago
1-step.yml	Initial commit	10 hours ago
2-step.yml	Initial commit	10 hours ago
3-step.yml	Initial commit	10 hours ago
4-step.yml	Initial commit	10 hours ago
python-coverage.yml	Added Coverage Workflow(Task 4)	44 minutes ago
python-coverage.yml.example	Initial commit	10 hours ago
python-package.yml	Added Test Workflow(Task 3)	1 hour ago
python-package.yml.example	Initial commit	10 hours ago

Python-package.yml

This screenshot shows the 'python-package.yml' workflow file. The left sidebar highlights the file in the 'workflows' directory. The main content area displays the YAML code for the workflow, which includes steps for checking out code, setting up Python, installing dependencies, running tests with flake8 and pytest, and linting with flake8.

```
1 name: Python package
2
3 on:
4   pull_request:
5     branches: [ "main" ]
6
7 jobs:
8   build:
9
10    runs-on: ubuntu-latest
11    strategy:
12      fail-fast: false
13      matrix:
14        python-version: ["3.9", "3.10", "3.11"]
15
16    steps:
17      - uses: actions/checkout@v4
18      - name: Set up Python ${{ matrix.python-version }}
19        uses: actions/setup-python@v5
20        with:
21          python-version: ${{ matrix.python-version }}
22      - name: Install dependencies
23        run: |
24          python -m pip install --upgrade pip
25          python -m pip install flake8 pytest
26          if [ -f requirements.txt ]; then pip install -r requirements.txt; fi
27      - name: Lint with flake8
28        run: |
29          # stop the build if there are Python syntax errors or undefined names
30          flake8 . --count --select=E9,F63,F7,F82 --show-source --statistics
31          # exit-zero treats all errors as warnings. The GitHub editor is 127 chars wide
32          flake8 . --count --exit-zero --max-complexity=10 --max-line-length=127 --statistics
33      - name: Test with pytest
34        run: |
35          pytest --verbose
```

Python-coverage.yml

This screenshot shows the 'python-coverage.yml' workflow file. The left sidebar highlights the file in the 'workflows' directory. The main content area displays the YAML code for the workflow, which includes steps for checking out code, setting up Python, installing dependencies, running tests with pytest and coverage, and generating a coverage comment.

```
1 name: Python Coverage
2
3 on:
4   pull_request:
5     branches:
6       - main
7
8 permissions:
9   pull-requests: write
10
11 jobs:
12   python-coverage:
13     runs-on: ubuntu-latest
14
15     steps:
16       - name: Checkout code
17         uses: actions/checkout@v4
18
19       - name: Set up Python
20         uses: actions/setup-python@v4
21         with:
22           python-version: 3.11
23
24       - name: Install dependencies
25         run: |
26           pip install -r requirements.txt
27           pip install pytest==8.4.1
28           pip install coverage==7.9
29           pip install pytest-cov==6.2.1
30
31       - name: Run tests and generate coverage details
32         run: pytest --cov=src
33
34       - name: Coverage comment
35         uses: py-cov-action/python-coverage-comment-action@v3
36         with:
37           GITHUB_TOKEN: ${{ github.token }}
38           MINIMUM_GREEN: 90
```